TrashScan: Design Report

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Introduction

Many restaurants throw significant quantities of reusable metal utensils into the trash every year, resulting in considerable economic losses, waste of resources, and safety risks for waste management. In response to this problem, our team developed TrashScan—a smart trash can prototype engineered to detect, retrieve, and sanitize metal components. Unlike conventional recycling or automated waste separation systems, TrashScan has a range of advanced features such as a copper induction loop sensor, LED indicators for real-time system feedback, UV sanitization to ensure germ-free recovered items, and a stepper motor combined with a pulley system to retrieve metal objects.

In our initial research phase, we studied existing metal detection techniques and observed that many designs use copper induction loops. Therefore, we designed TrashScan to use a tuned copper induction loop made with 0.9 mm enameled copper wire. The loop generates an alternating magnetic field and, when a metal object enters the field, it creates eddy currents that cause a measurable shift in the voltage decay. This operational principle—related to Faraday's law of induction—is documented in the literature [1].

Furthermore, in the mechanical design, we selected a stepper motor driving a pulley system based on a block-and-tackle configuration. Research shows that these systems can reduce the necessary lifting force compared to direct lifting, which is essential when using a lightweight motor in a repetitive cycle [3].

An obstacle we encountered during early testing was the recurring triggers of the induction loop detection, even after the metal was picked up. To fix this, we integrated a system that stops detection and asks the user for a button press after the metal is retrieved. This way we override that deactivate the sensor temporarily and once an operator confirms the retrieval, TrashScan goes into detection mode again [2].

Lastly, the entire system is managed using an Arduino microcontroller. Using Arduino's open-source platform helped us prototype efficiently, and also manage user input. I played a critical role not only in assembling the electronics but also in integrating them into a custom-designed 3D-printed enclosure. Although I did not create the enclosure personally, I helped assemble all the components inside it, ensuring that the design is both robust and aesthetic.

*Figure 1: **Early prototype** layout displaying sensor and LED mounting configuration.*

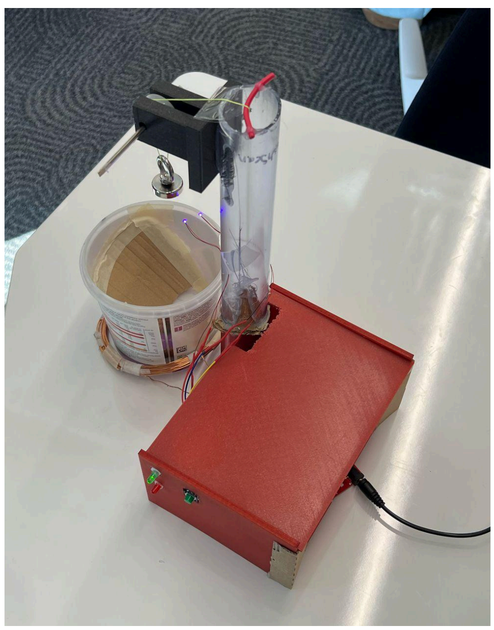

*Figure 2: **Current Prototype** with 3D printed case*

Design

Electronic and Sensor Systems

I assembled the electronic components and developed the firmware to control the system. TrashScan's sensor system is a copper induction loop made with 0.9 mm enameled copper wire. The loop works by generating an alternating magnetic field which, according to Faraday's law, will produce an induced voltage (E) that is proportional to the rate of change of the magnetic flux (dΦ/dt) through the coil. This relationship is expressed mathematically as

$$E = -N(\frac{d\Phi}{dt})$$

where N is the number of turns of the coil. When a metal object enters the field, eddy currents are induced within it, thereby altering the local magnetic flux. This effect leads to a noticeable shift in voltage decay, which our sensor circuitry detects [1]. In practice, calibrating the sensor to detect the voltage shift accurately is necessary to detect metal. By examining the decay pattern of the induced signal, our system discriminates between noise and a genuine metal object.

To further increase reliability, I programmed a counter algorithm into the Arduino-based control system. The algorithm continuously monitors the sensor readings and confirms the presence of metal only when five consecutive positive signals are detected, each spaced by 500 milliseconds. This multi-read confirmation technique has been effective in reducing false positives caused by sudden changes in ambient conditions, as noted in prior works on inductive sensor accuracy [1].

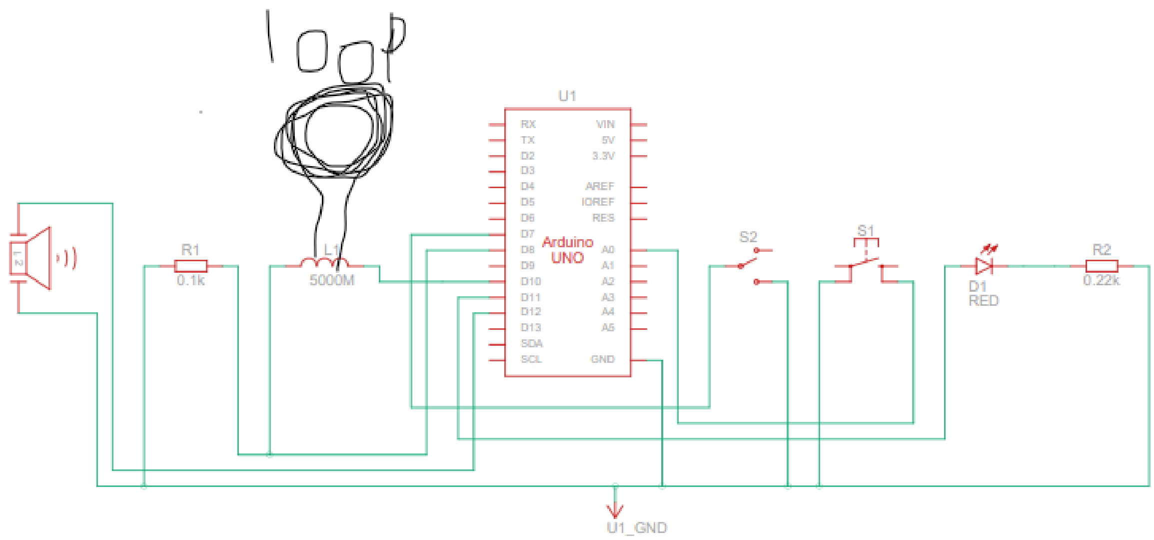

*Figure 3: **Early prototype** schematic diagram Schematic diagram of TrashScan*

Mechanical Retrieval System

After sensor confirmation, the system activates a stepper motor for the retrieval mechanism. For TrashScan, I selected a stepper motor due to its ability to offer precise control over movement, which is critical for accurately positioning the retrieval mechanism. However, the motor's torque was not sufficient for repeated and reliable metal lifting. To address this limitation, I researched various pulley systems and ultimately incorporated a simpler version of the block-and-tackle design into the mechanism.

A block-and-tackle setup uses multiple pulleys to distribute the load, thereby reducing the effort required by the motor [3]. I connected the pulley *through* the magnet and glued it to the stationary wall. This essentially divided the weight in half.

Sanitizing UV Lights

When the pulley returns with the load attached to it, it turns on Ultra Violet (UV) lights to disinfect not only the magnet hook, but also the magnetic load attached. This idea was not mine, it was from one of my teammates and we all agreed that it added a great extra function to TrashScan.

Arduino Microcontroller Integration

Central to the operation of TrashScan is the Arduino microcontroller, which serves as the brain of the system. Moreover, all the electronics and wiring were neatly assembled into a 3D-printed enclosure. Although I did not perform the 3D printing myself, I contributed significantly to the internal layout and wiring configuration to ensure that all components were securely housed and that the overall design was both durable and aesthetically pleasing.

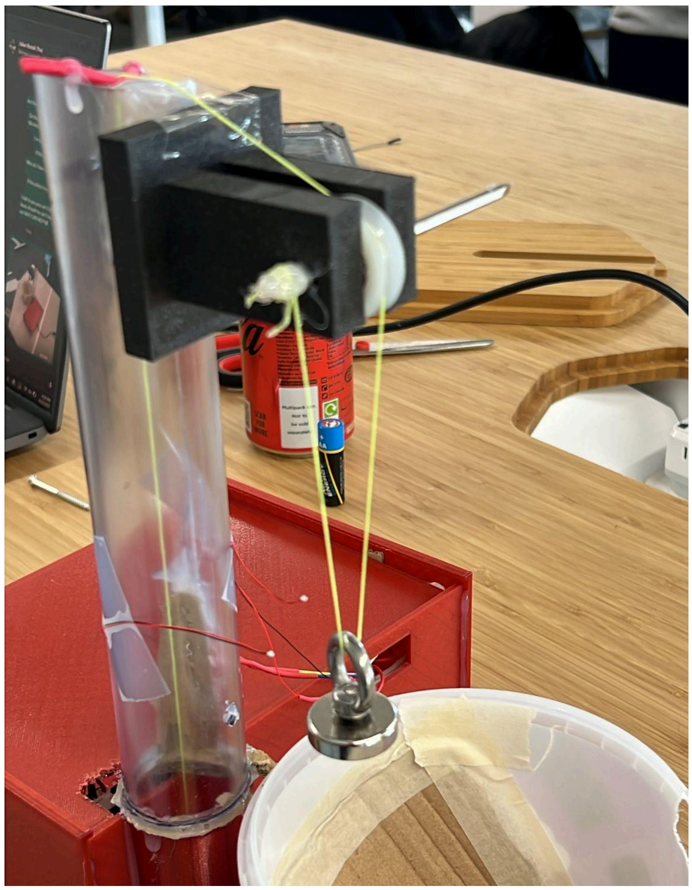

Figure 4: Stepper motor and pulley system assembled in a block-and-tackle configuration to reduce mechanical force requirements

```
// Skip detection while in stepper mode
if (!detection_mode) continue;

// Measure
long int val = meas(period, minlen, maxlen, steplen, nloop);

// Calculate
int diff = val - ref;
absdiff = abs(diff);

// Check for metal detection condition
if (absdiff > threshold * 12) {
  metal_detected = true;
  consecutive_count++; // Increment consecutive counter
```

Figure 5: Arduino code snippet demonstrating sensor integration and detection confirmation logic

Video 1: TrashScan demonstration highlighting detection, retrieval, and manual override operations, and UV lights ([CLICK](#) TO PLAY)

Iterative Testing, Calibration, and Failure Mode Analysis

A critical phase of development was the iterative testing and calibration of both the sensor and mechanical systems. While validating the induction loop sensor, I encountered persistent issues. The initial loop (made with 0.9 mm wire) was prone to false readings. I experimented with an alternative loop made from 0.5 mm wire. Unfortunately, this thinner wire produced no output at all, which led to the discovery that the fault was not with the sensor's design but with a wiring error during assembly.

Along with the sensor issues, early tests with the direct motor lift mechanism revealed that the stepper motor was not capable of pulling up the load effectively. The lack of sufficient force led us to adopt a simple version of the block-and-tackle pulley system, which successfully provided the necessary mechanical advantage. This adjustment significantly improved the motor's performance in lifting metal utensils.

Moreover, false detections were observed during initial trials. The sensor occasionally caused fake triggers, resulting in the stepper motor engaging repeatedly despite the metal object already being retrieved. To correct this, I created the counter algorithm to require multiple consistent positive readings before motor activation.

One last obstacle we had was that TrashScan, after retrieving a metal, kept on reading and retrieving. This was bad because the loop would still detect the metal connected to the pulley system. Thus, I had to integrate into the system a button to be pushed once the metal was in the air connected to the pulley system. When the button is pressed, the loop starts reading again and metal can be detected.

While iterating, my teammate had the idea to add a sanitization as a new feature to the product.

These iterative adjustments and troubleshooting steps were necessary in taking TrashScan from a conceptual prototype to a functional prototype capable of operating reliably in real-world environments [4]. These hands-on modifications not only enhanced system reliability but also deepened my understanding of the connection between hardware and software in embedded system design.

Summary and Conclusion

TrashScan was made as a solution to the significant problem of metal silverware waste in restaurant settings. By integrating a copper induction loop sensor with a stepper motor-driven block-and-tackle pulley system and controlling the entire operation via an Arduino microcontroller, we developed a system capable of detecting, retrieving, and sanitizing metal objects efficiently. The system's design is distinguished by its use of a counter algorithm—requiring five consecutive positive sensor readings at 500-millisecond intervals—to reduce false detections, and by the addition of a manual push-button override to halt redundant cycles.

Through extensive iterative testing and calibration, I was able to determine the optimal mechanical parameters, including the ideal motor rotations, string length, and sensing parameters, ensuring that the system operates reliably and safely. The systematic process of prototyping, testing, and refining components provided valuable practical insights that will guide future improvements.

This project underscores the importance of integrating theoretical principles with hands-on experimentation. By applying well-established concepts in electromagnetic induction [1], mechanical advantage through pulley systems [3], and rapid prototyping with Arduino [5], TrashScan represents a successful synthesis of theory and practice. The experience has not only expanded my technical skills but also deepened my understanding of system integration and iterative design methodologies. Going forward, I plan to further refine the system to enhance its performance and reliability, making it a viable solution for sustainable waste management in restaurant environments.

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Ethical Evaluation of TrashScan Project

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Introduction

In our engineering project, we have drawn upon several major ethical theories to guide and evaluate the morality of our project. Concepts from consequentialism (Utilitarianism) [1][5][6] help us assess whether our actions result in the greatest overall good for all affected stakeholders. Deontological ethics [1][6] focuses our attention on upholding duties and moral rules, ensuring that our actions respect the rights and obligations inherent in the engineering process. Meanwhile, virtue ethics [1][6] centers on our personal character—evaluating whether our design and implementation express virtues such as honesty, care, and responsibility. In addition, we use value theory [7] to consider what should be prioritized in our decisions, such as the balance between human welfare and environmental health. Together, these theories provide us with complementary viewpoints that inform how we evaluated our TrashScan project.

Our team developed TrashScan as a smart trash can prototype to address a significant real-world waste problem. We observed that restaurants accidentally discard large quantities of reusable metal silverware—often amounting to thousands of pieces per year per establishment [4]—which not only results in economic loss and resource waste but also poses safety risks for those handling the waste. With TrashScan, we sought to detect and separate metal utensils from the general waste stream and then sanitize them with ultraviolet (UV) light [2] so that valuable silverware is not simply discarded. In this report, we examine how our engineering solution addresses key concerns such as environmental impact, worker health and safety, cost efficiency, and improved hygiene, using the lenses of consequentialist, deontological, and virtue ethics frameworks as well as a broader value-based approach [5]–[7].

Ethical Analysis of TrashScan's Impacts

Consequentialist (Utilitarian) Perspective

From our utilitarian standpoint, we consider TrashScan to be ethical because it produces significant net benefits for multiple stakeholders. The primary measure of success is whether the positive outcomes of our design outweigh any potential costs or harms. In our case, the environmental benefits are clear. By recovering metal utensils from waste, TrashScan helps ensure that fewer reusable items end up in landfills. This not only conserves natural resources but also saves energy; “For example, using recycled aluminum cans to make new aluminum cans uses 95% less energy than using bauxite ore, the raw material aluminum is made from” [3]. By preventing unintentional silverware loss, our project helps reduce the pollution and carbon emissions that occur when manufacturers produce replacement items, thereby supporting environmental sustainability. Although these benefits may be indirect, they contribute positively to society by preserving resources and reducing greenhouse gas emissions.

Moreover, TrashScan creates economic benefits. Restaurants lose money by unintentionally throwing away silverware—studies indicate that an average restaurant might lose around 17 pieces of silverware per day, resulting in thousands of dollars in losses each year [4]. Our prototype detects and recovers these utensils, allowing them to be cleaned and returned to use, thereby reducing the need for purchasing new items. This cost-saving measure not only benefits the business directly but also allows saved resources to be reinvested in other areas.

In addition, there are important health and safety improvements. By separating sharp or hazardous metal objects from the waste stream, TrashScan reduces the risk of injury to sanitation workers who otherwise might be exposed to dangerous or contaminated items. In many cases, restaurant staff have had to rummage through trash in search of silverware [8]—a task that is both unpleasant and unhygienic. Our design minimizes such hazards and provides an additional hygiene safeguard with its UV sanitation feature [2], which kills bacteria and viruses on the retrieved utensils. These measures not only improve working conditions for waste handlers but also enhance public health by reducing disease transmission.

Deontological (Duty-Based) Perspective

Evaluating TrashScan from a deontological perspective means focusing on our responsibilities and the moral imperatives that guide our work, regardless of the outcomes. One of our most fundamental ethical duties is to prevent harm to others whenever possible. We recognize that improperly disposing of metal utensils can create hazards—such as injury to a waste worker—which we consider an avoidable harm. By incorporating TrashScan into our waste management process, we fulfill our obligation to ensure that our operations do not inadvertently endanger others. Our project thereby upholds the principle of “do not harm,” which is central to deontological ethics [1][6].

Furthermore, certain actions are morally required regardless of their consequences. For us, ensuring that waste is handled safely and hygienically is a required duty. The UV sanitation component of TrashScan [2], for example, embodies our commitment to properly managing waste. By sterilizing discarded items, we reduce the risk of disease and honor our responsibility to both our workers and our community. In this way, TrashScan supports a framework that values moral duty over outcome calculation, making it an ethical choice.

Virtue Ethics Perspective

The virtue ethics approach shifts the focus from specific rules or outcomes to the character and intentions behind our actions. When developing TrashScan, our team aimed to embody virtues such as responsibility and ingenuity[1][6]. Our choice to tackle the problem of wasted silverware and unsanitary waste shows a sense of social responsibility and moral attentiveness in engineering. In designing TrashScan, we demonstrated practical wisdom—balancing environmental concerns, economic realities, and public health needs in a single integrated solution.

For us, creating a system that not only recovers valuable utensils but also ensures they are safe for reuse shows our commitment to ethical engineering practice. We are demonstrating strong ethical character by putting our effort into a system that addresses these multiple needs. These include the consideration of the risks to waste management workers [8] or the inclusion of a UV sanitation feature [2], after all, every aspect of TrashScan shows our dedication to upholding value principles in engineering design.

Value Theory and Stakeholder Values

Beyond the formal ethical theories, it is crucial to consider the value-based framework that is in our project [7]. Our design priorities include environmental sustainability, public health, and economic efficiency. In developing TrashScan, we have consciously balanced a range of stakeholder values: the economic interests of restaurant owners [4], the safety and well-being of sanitation workers [8], the environmental benefits of recycling and reusing materials [3], and the broader public health implications of improved waste management [2].

Our approach is value-sensitive. We ensure that the financial benefits—such as recovering lost silverware [4]—are not achieved at the expense of environmental and social considerations. Instead, our project addresses multiple values at the same time. For instance, recovering and sanitizing metal utensils [2] not only conserves resources and saves money but also contributes to a reduced environmental footprint by lessening the need for new manufacturing [3]. Including stakeholder values exemplifies a design philosophy that treats the environment, human welfare, and economic efficiency as interconnected rather than mutually exclusive elements.

Summary and Conclusion

Evaluating the TrashScan project through these various ethical lenses has allowed us to gain an understanding of its moral dimensions. The consequentialist analysis [1][5] clearly shows that our design maximizes overall benefits by promoting environmental sustainability [3], achieving cost savings [4], preventing injuries [8], and enhancing public health [2]. Although there are some costs involved in the implementation of our prototype, the overall utility produced by these benefits justifies our approach.

The deontological analysis [1][6] reinforces that by adopting TrashScan, we are fulfilling our moral duty to prevent harm and maintain a safe and hygienic waste management process—an obligation that we believe is important in any responsible engineering practice.

Finally, by applying value theory [7], we have ensured that our project respects a broad spectrum of stakeholder interests. Our inclusive approach means that TrashScan not only provides economic benefits to restaurant owners [4] but also supports the safety of workers [8] and environmental sustainability [3], making it a well-rounded solution from both an ethical and technical standpoint.

In conclusion, our TrashScan project stands as a testament to how ethical principles [1], [5]–[7] can be integrated into innovative engineering design. By addressing the practical problem of wasted reusable utensils [4] and enhancing waste management practices through safety and hygiene improvements [2], [8], our prototype aligns with the ideals of utilitarianism, deontology, and virtue ethics. Ultimately, our work demonstrates that engineering innovation can—and should—be both technically successful and morally sound, paving the way for future improvements in sustainable design.

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